

0005 — RevFT Trade Location: VWAP Deviation & Value Area — 2026-06-24

****Series:**** MC Setup Research Notes · Note 0005 ****Confidence:**** High ****as a negative result**** — 5 years, full-cost, real-tick, the headline gate was ****pre-registered**** and judged in R with year-by-year CIs. We are confident there is no tradeable location edge here; we are not claiming a positive system.

TL;DR: We tried to rescue the RevFT ("MyReversals") signal set — a firm net loser at 1:1 — by finding where it should be traded. Three location axes: distance to the day's developing extremes (prior work), deviation from the developing-session VWAP, and position vs the prior-day value area. **The obvious intuition — that a reversal works better when price is stretched far from VWAP — is not just unsupported, it is backwards: the deeper the stretch, the worse the fade (it gets run over).** The only coherent signal points the other way: RevFT behaves weakly like a with-trend continuation trade (better on the far side of VWAP in the trade's direction, improving with a wider target). But when we pre-registered that as a gate and tested it honestly, it only reaches **break-even**, and its entire result is carried by a single year (2022); 2025 and 2026 are significantly negative. **Recommendation: there is no location fix for RevFT. Retire it as a fade. Its problem is directional/structural, not where you take the trade.**

1. The setup (so this note stands alone)

The RevFT trade. "RevFT" (a.k.a. MyReversals) is a swing-reversal signal on 5-minute ES: it fires when price rejects a recent swing extreme, betting on a turn. You enter at the first tick after the signal bar closes, stop just beyond the rejected extreme (StopPrice), and exit at a fixed multiple of **R** (R = entry-to-stop distance). ES: 1 point = \$50/contract, 1 tick = \$12.50. Two price anchors are recorded per signal: SignalPrice (the trigger, a few ticks back inside the swing) and StopPrice (just beyond the extreme the reversal formed off).

Why we're even here. At a 1:1 target RevFT loses money outright — **-\$187k net / -\$79k even frictionless**, 6,133 trades over five years (Note: this is a real directional loss, not pure cost bleed). Earlier work showed you cannot fade or flip your way out of it. The remaining hope was **selection**: maybe a subset, defined by trade location, clears the ~\$15/trade cost hurdle. This note tests that hope and closes it.

The two location features (both look-ahead-safe).

- **VWAP deviation (VWAP_dev).** The developing-session VWAP is the volume-weighted average price so far today (resets each RTH session, uses only bars up to the signal). $VWAP_dev = (Close - VWAP) / \sigma$, where σ is the volume-weighted standard deviation of price from VWAP so far. It is the signed **number of σ** the signal sits from VWAP: negative = below VWAP, positive = above. For a **long**, a mean-reversion fade (buy an oversold dip) is **below** VWAP ($VWAP_dev < 0$); for a **short**, it is **above** ($VWAP_dev > 0$).
- **Prior-day value area (VAH / VAL).** The price band containing 70% of yesterday's volume (POC = point of control, VAH/VAL = its high/low). We use the **prior completed** session's value area as today's reference — never today's developing one — so it is causal. vaD_loc = signal above / inside / below it; vaD_dist = signed distance beyond the edge.

Cost model. \$4.36 round-turn commission + 1 tick entry slippage, 0 exit. All sims replay **real continuous ticks day-by-day** through the production engine. Signal set: `ba_signals_revft.parquet` (6,328 signals, 2021-06 → 2026-06; 6,133 fill). Features are tagged **causally** through the S34-corrected `tag_signals` chokepoint (see Note 0002 for the look-ahead fix this depends on).

2. The question

1. **Is there a subset of RevFT, defined by where the signal sits relative to VWAP and to value, that turns the loser into a winner — or at least clears the cost hurdle?** 2. **Specifically: do reversals work better when price is extended from VWAP (the standard mean-reversion intuition)?** 3. **If a pattern appears, is it a real edge or a single-regime artifact?** (Pre-register the gate; judge in R; check every year.)

3. How we tested it (the menu)

Three descriptive bucket studies, each on the real tick engine, pinned single-leg, with $\pm 95\%$ CIs, judged in **R** (not dollars — see §4.6 on why):

1. **Static location** (prior work, recapped): signed distance from the **developing HOD/LOD** and **prior-year HOY/LOY**, both anchors.
2. **VWAP deviation:** bucket filled trades by $VWAP_dev$. First a coarse directional pass, then **0.5σ slices, long-only and short-only separately** (no sign transform — read straight off the table), at **1R / 2R / 3R** to test whether a tight exit was hiding a reversion edge.
3. **Prior-day value area:** categorical location (above / inside / below) and signed distance beyond the responsive-fade edge.
4. **Confirmation:** the one coherent pattern (a continuation tilt) was written down as a **pre-registered gate** and tested full-sample + year-by-year at 1R/2R/3R.

Engine: tick-by-tick first-touch (entry at first tick after the signal; stop/target by tick crossing), unlimited-positions scoring.

4. Results

4.1 Static location (prior work) — already dead

Bucketing by distance to the **developing day extreme** (the "catch the knife at the low/high" idea): the at-the-extreme band (± 0.05 ADR) is the **worst** bucket, with no gradient toward it.

Distance to developing extreme (ADR, SignalPrice)	n	Exp R	PF	Win %
-0.05 .. 0.05 (AT the extreme)	74	-0.296	0.68	36.5%
0.05 .. 0.15	1,622	-0.130	0.81	44.3%
0.15 .. 0.30	2,001	-0.056	0.94	47.5%
> 0.60 (deep inside range)	782	-0.034	0.88	48.1%

Firing right at a still-forming extreme = catching a knife into a level still being made. Prior-year HOY/LOY proximity was equally null. **Static location: closed.**

4.2 VWAP deviation, directional view — no edge, hint of the wrong sign

Re-expressing VWAP_dev as "stretch on the mean-reversion-favorable side" ($S = -dev$ for longs, $+dev$ for shorts; $S > 0$ = the textbook fade location), 1R, both directions:

Directional stretch S (σ)	n	Exp R	PF	Win %
< -1.0 (fading on the continuation side)	1,368	-0.052	0.95	47.6%
-0.25 .. 0.25 (at VWAP)	675	-0.055	0.93	47.7%
1.0 .. 2.0 (favorable stretch)	1,550	-0.087	0.88	46.3%
2.0 .. 3.0 (deep favorable stretch)	54	-0.566	0.28	22.2%

Every bucket loses, and the **most "favorable" (deep) stretch is catastrophic**. The gradient runs the wrong way. This coarse view flagged the result; the slices below nail it.

4.3 The decisive test — 0.5σ slices, no sign transform (1R)

Raw VWAP_dev, long-only and short-only, so there is zero sign ambiguity. For a **long**, the mean-reversion fade is the **left** (negative) cells; for a **short**, the **right** (positive) cells.

LONG only (1R):

VWAP_dev (σ)	n	Exp R	$\pm R$ CI	PF	Win %
-2.5 .. -2.0 (deep fade)	15	-0.479	± 0.464	0.35	26.7%
-2.0 .. -1.5	170	-0.107	± 0.143	0.84	45.3%
-1.5 .. -1.0	522	-0.052	± 0.082	0.98	48.9%
-1.0 .. -0.5	498	-0.107	± 0.084	0.84	45.8%
-0.5 .. 0.0	389	-0.132	± 0.093	0.77	43.2%
0.0 .. +0.5	306	-0.050	± 0.106	0.95	47.4%
+1.0 .. +1.5 (continuation)	412	+0.005	± 0.091	1.17	49.8%
+2.0 .. +2.5	74	+0.016	± 0.213	1.17	52.7%

SHORT only (1R):

VWAP_dev (σ)	n	Exp R	$\pm R$ CI	PF	Win %
-3.0 .. -2.5 (continuation)	30	+0.197	± 0.344	2.28	60.0%
-2.5 .. -2.0	83	-0.099	± 0.203	0.84	44.6%
-0.5 .. 0.0	321	-0.055	± 0.105	0.77	47.7%
+0.5 .. +1.0	484	-0.075	± 0.085	0.91	46.5%
+1.0 .. +1.5 (favorable fade)	568	-0.062	± 0.079	0.91	47.0%
+1.5 .. +2.0	290	-0.188	± 0.111	0.63	41.0%
+2.0 .. +2.5 (deep favorable fade)	38	-0.641	± 0.250	0.23	18.4%

Read it directly: **fading into a $2\sigma+$ VWAP stretch (long far below / short far above) is the worst thing you can do** — the extension keeps going and runs over the fade. The only green cells are on the **continuation** side (long above VWAP, short below). RevFT signals are also nearly symmetric around VWAP (longs median -0.10σ , shorts $+0.37\sigma$) — they mostly fire near VWAP, so "extended reversal" is a small, hostile corner.

4.4 Was a tight 1:1 hiding it? No — 2R and 3R say the same

If the reversion edge were real but clipped at 1R, widening the target would reveal it. It doesn't — the reversion (deep-fade) cells stay red at every target, while the continuation cells get better with a wider target (a momentum signature). Same two cells across targets:

Cell	1R	2R	3R
Long deep fade ($-2.5..-2.0\sigma$)	-0.479	-0.319	-0.452
Short deep fade ($+2.0..+2.5\sigma$)	-0.641	-0.573	-0.588
Long continuation ($+1.0..+1.5\sigma$)	+0.005	+0.032	+0.091
Short continuation ($-3.0..-2.5\sigma$)	+0.197	+0.393	+0.505

(Full 14-band × long/short × 1R/2R/3R grids in the saved artifact.) The exit is not the problem. The direction of the premise is.

4.5 Prior-day value area — same story, no independent edge

Categorical location (1R, both directions):

Signal vs prior value area	n	Exp R	PF	Win %
above prior VAH	2,287	−0.067	0.95	46.9%
inside	1,897	−0.093	0.87	46.0%
below prior VAL	1,949	−0.096	0.83	45.4%

"Above value" is least-bad — again the continuation/breakout side, not the responsive-fade side. The directional "distance beyond the responsive-fade edge" (V) trends the wrong way too: the **deeper beyond the value edge, the worse** ($V > 0.60 \rightarrow -0.134R$). No value-area cell clears the hurdle with a CI excluding zero.

4.6 The continuation inversion — pre-registered, then judged honestly

Every coherent signal pointed one way: **RevFT works (weakly) with the trend, not against it**. We wrote that down before testing as a fixed gate:

continuation = (Long & VWAP_dev > +0.5σ) OR (Short & VWAP_dev < −0.5σ)

and judged it in R, full sample, at three targets. A caution first: the apparently positive dollar pockets in \$4.5/\$4.2 are a **wide-stop illusion** — they win in dollars only because their trades carry bigger stops (more \$ per R), while their R is flat-to-negative. The Keystone note hit the same trap; we judge in R for exactly this reason.

Target	CONTINUATION gate	R 95%-interval	REVERSION complement	NEUTRAL (dev ≤0.5)
1R	−0.072	[−0.113, −0.031] loser	−0.099	−0.081
2R	−0.034	[−0.088, +0.021] incl. 0	−0.108	−0.113
3R	−0.016	[−0.078, +0.046] incl. 0	−0.104	−0.074

Two honest readings:

- **The split is real and robust.** Continuation beats reversion by ~0.07–0.09R at every target, and the reversion complement's interval excludes zero on the negative side throughout. RevFT is structurally a continuation signal, not a fade.
- **But continuation alone only reaches break-even.** It climbs −0.072 → −0.034 → −0.016R as the target widens and asymptotes at zero — it never becomes positive-significant. Avoiding the loss is not making money.

4.7 Year-by-year — the gate is one regime, not an edge

The continuation gate at 3R, by year — the test that decides it:

2021	2022	2023	2024	2025	2026
−0.149 (incl. 0)	+0.185 [+0.035,+0.336]	−0.037 (incl. 0)	+0.081 (incl. 0)	−0.150 [−0.274,−0.025]	−0.203 [−0.389,−0.017]

The entire break-even result is **carried by 2022** (+\$45,650 on one contract — a strongly trending bear year where continuation paid). Strip 2022 and the gate is a loser, with **2025 and 2026 significantly negative**. This is the textbook one-regime artifact. Pre-registering the gate and demanding year-by-year stability is precisely what stops this from being mis-logged as a "continuation edge."

5. Why it fails

Why the fade premise is wrong. A VWAP that price has stretched 2σ away from is not an exhausted rubber band on this signal — it is an imbalance in progress. RevFT's swing-rejection trigger fires into that imbalance, and the imbalance wins more often than it reverts. Fading deeper extension just buys a worse spot in a continuing move. That the deepest-stretch cells are the most negative (long $-2.5\sigma \rightarrow -0.48R$; short $+2.0\sigma \rightarrow -0.64R$) is the clean fingerprint of "trend, not mean-reversion."

Why even the continuation inversion isn't an edge. Selection can only concentrate structure that exists; it cannot manufacture it. RevFT's base population has a real negative directional edge (it loses even frictionless). Slicing it by VWAP side finds the least-negative corner — the with-trend trades — and that corner happens to be ~zero on average and strongly positive only when the whole market trends (2022). It is regime beta, not a stable selection edge. No exit (1R–3R) and no location (extremes, VWAP, value) repairs a signal whose core directional sign is wrong.

6. Recommendation

1. There is no trade-location fix for RevFT. Retire it as a fade. Every location axis is dead: static extremes (§4.1), VWAP deviation (§4.2–4.4), prior-day value area (§4.5). The mean-reversion intuition is falsified, not merely unconfirmed — extended fades are the worst trades in the book.

2. Do not deploy the "continuation" version either. It is break-even at best (§4.6) and its positive years are one regime (2022); 2025–26 are significantly negative (§4.7). It would fail forward.

3. The only durable takeaway: if RevFT is ever used at all, it must be **with-trend** (on the far side of VWAP, in the trade's direction) and **never as a fade into extension** — but treat even that as a 2022 artifact until proven on truly held-out data. The honest conclusion is that RevFT's problem is **directional/structural, not locational**, and effort is better spent elsewhere (e.g. the audited MC IB-edge fade, Note 0003).

7. Caveats & open questions

- **This is a negative result, stated with confidence.** Five years, full cost, real ticks, pre-registered gate, judged in R with CIs — strong evidence of absence of a location edge. It is not proof that no conceivable feature rescues RevFT, only that the two the structure most strongly suggested (VWAP stretch, value-area edge) do not.
- **In-sample thresholds.** The $+0.5\sigma$ continuation cut was chosen on this same 5-year set; §4.7 shows it fails year-by-year anyway, which only strengthens the negative.
- **The continuation tilt is real but small and regime-bound** — worth remembering as a property of the signal ($\text{RevFT} \approx$ momentum, not reversion), not as a tradeable rule.
- **Developing (intraday) value area untested.** We used the prior-day value area (the look-ahead-safe one). A causal developing VA-at-the-bar is buildable but, given how decisively both the VWAP and prior-VA axes failed, is low-priority.
- **Selectivity-at-wider-R for the whole set** (trade far less, not by location) remains formally open, but two independent location studies plus a falsified premise make it a poor bet.

8. Reproduce

Scripts (headless, day-by-day tick replay against the production engine; no production code changed):

- `scripts/reversal_at_extreme_study.py` — static location: developing HOD/LOD & HOY/LOY buckets (§4.1)
- `scripts/revft_vwap_va_location.py` — directional VWAP stretch, prior-day value area, 2-D cross (§4.2, §4.5)
- `scripts/revft_vwap_slices.py` — 0.5σ VWAP_dev slices, long/short, at 1R/2R/3R + sign sanity (§4.3, §4.4)
- `scripts/revft_continuation_gate.py` — pre-registered continuation gate, full-sample + year-by-year (§4.6, §4.7)

Saved artifacts (in `docs/living/`): `reversal_at_extreme_study_20260624.md`, `revft_vwap_va_location_20260624.md`, `revft_vwap_slices_20260624.md`, `revft_continuation_gate_20260624.md`.

Feature definitions (look-ahead-safe, via `tag_signals`): $\text{VWAP_dev} = (\text{Close} - \text{VWAP})/\sigma$ on the developing session VWAP (`auction_features.session_vwap_bands`); prior-day value area `vaD_VAH/VAL/loc/dist` from the prior completed session's 70%-volume profile. Continuation gate: (Long & $\text{VWAP_dev} > +0.5$) OR (Short & $\text{VWAP_dev} < -0.5$), judged in R at 1R/2R/3R.